# Aleksandr Medvedev

**RAG | NLP | PyTorch | TensorFlow**

Moscow | 30+ h/week | medv.ax@outlook.com | github.com/MedvAx-AI | medvax-ai.github.io

## SUMMARY

3rd-year Applied Artificial Intelligence student building LLM and AI-agent systems with RAG, NLP, PyTorch, TensorFlow, and production ML tooling. Built an auditable routing system evaluated across 1,000 Qwen2.5 runs on 500 MMLU questions that reduced p50 latency by 15% while adding human review and traceable RAG controls. Focused on AI agents and multi-agent systems for operational risk, ESG risk, and applied risk analytics.

## SKILLS

**LLM & AI:** LLMs, AI agents, RAG, Hugging Face Transformers, FAISS, LLM evaluation, human-in-the-loop systems

**ML & NLP:** Python, PyTorch, TensorFlow, Scikit-learn, Pandas, NLP, text embeddings, semantic search

**Engineering:** Git, GitHub Actions, FastAPI, Docker, DVC, MLflow, PostgreSQL

**Focus Areas:** Operational risk analytics, ESG risk, multi-agent systems

## PROJECTS

### BudgetRoute-LLM - LLM Routing Agent with RAG & Human Review

Python | PyTorch | FastAPI | FAISS | PostgreSQL | github.com/MedvAx-AI/budgetroute-llm

- **Reduced p50 inference latency by 15% (390 ms vs 461 ms)** by routing 78% of requests to Qwen2.5-0.5B while selecting between small/large models, RAG retrieval, cascade, abstention, and human review.
- **Identified 9.8 pp of routing headroom (65.0% oracle vs 55.2% always-large)** across 1,000 Qwen2.5 runs by evaluating five routing strategies on 500 MMLU questions.
- **Achieved 1,000/1,000 reproducible answers on a clean rerun** by using leakage-free 300/100/100 splits, pinned model revisions, and bootstrap confidence intervals for held-out evaluation.
- **Improved auditability and operational-risk controls** by implementing a human-review queue, tamper-evident audit chain, traceable RAG evidence (chunk ID, score, source), and a PostgreSQL control plane.

### Yacht Resistance MLOps Pipeline

Scikit-learn | Pandas | DVC | MLflow | Docker | github.com/MedvAx-AI/pmldl-yacht-mlops

- **Achieved R2 0.993 and RMSE 1.28 on 70 observations from 5 unseen hulls** by training an Extra Trees pipeline with hull-grouped splits and automatic baseline validation.
- **Eliminated manual retraining and deployment steps** by automating DVC -> MLflow -> FastAPI/Streamlit every 5 minutes, verified by 10 CI tests and model SHA-256 plus MLflow run-ID checks.

### Beyond Proximity / SemanticSplat - Research Team Lead

- **Led a 4-person team to complete a 150-query evaluation across 5 scenes** by coordinating experimental execution plus ScanNet and Replica pilot studies.

## EDUCATION

**Innopolis University** | BSc Applied Artificial Intelligence | 2024-2028